WEEK 1

AWS ENGINEER SPRINT

Account Security • S3 Data Vault • Recovery • Network Security

Name	**Imtiyaaz Agmad Begg**
Programme	Sip & Cloud Tech – Work Integrated Learning
Week	2–6 November 2026
Submission date	06 November 2026

Week 1 practical work and evidence

1. Introduction

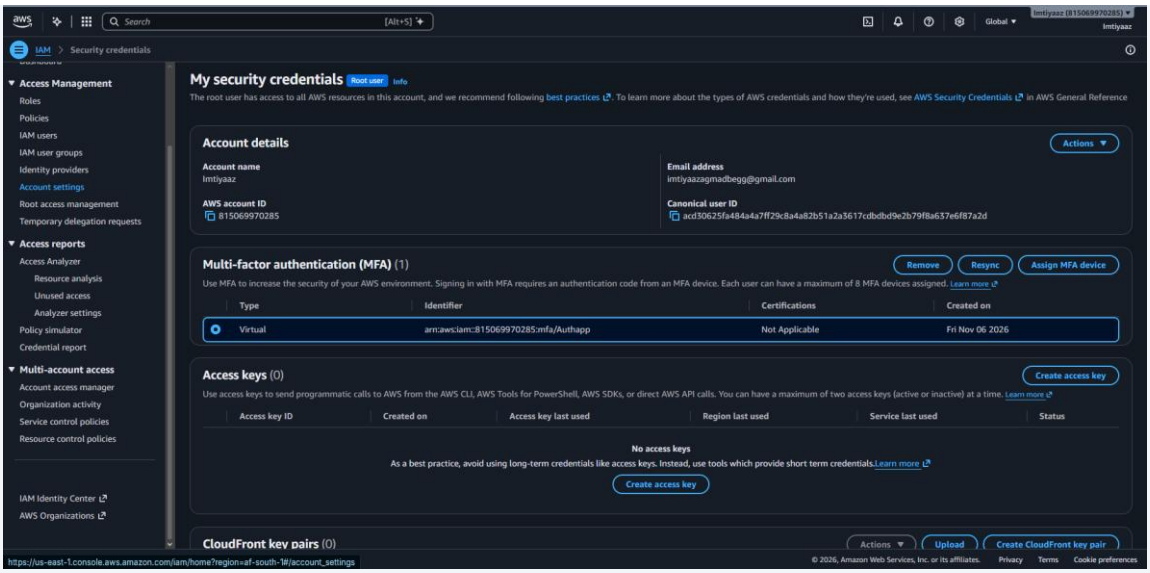
During Week 1, I worked on setting up and securing my own AWS environment. The main focus was getting the basic account controls in place, managing possible costs, creating a private S3 storage bucket and learning how to recover a file using versioning. I also worked on network security and the idea of only allowing access when it is actually needed.

This document explains what I worked on and includes spaces for the screenshots that prove each setting or task. I have kept the explanations straightforward so it is clear what I did and why it matters.

2. Day 1 – Account Security, Billing and Learning Platform

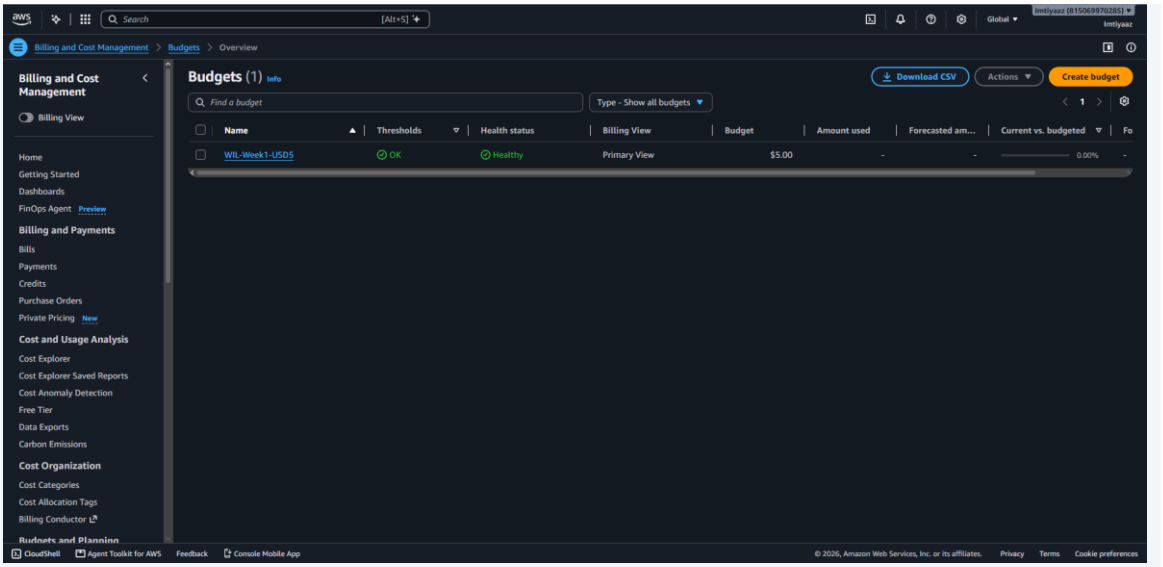
2.1 Root account MFA

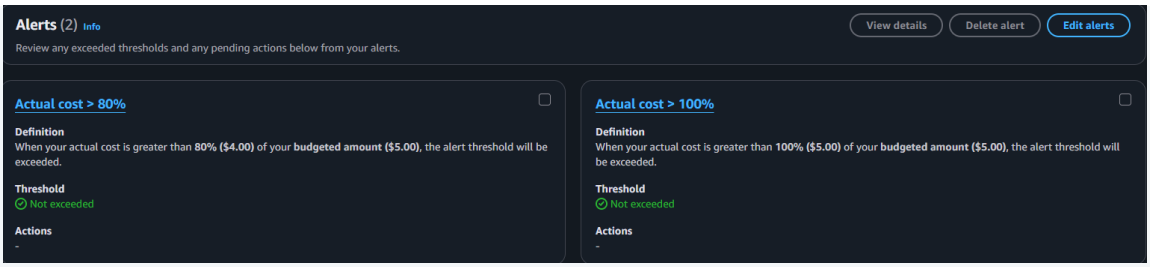
I checked the security settings on my AWS root account and confirmed that multi-factor authentication (MFA) was already enabled. MFA adds another verification step when signing in, which helps protect the account if the password is ever exposed.



2.2 Billing budget and alerts

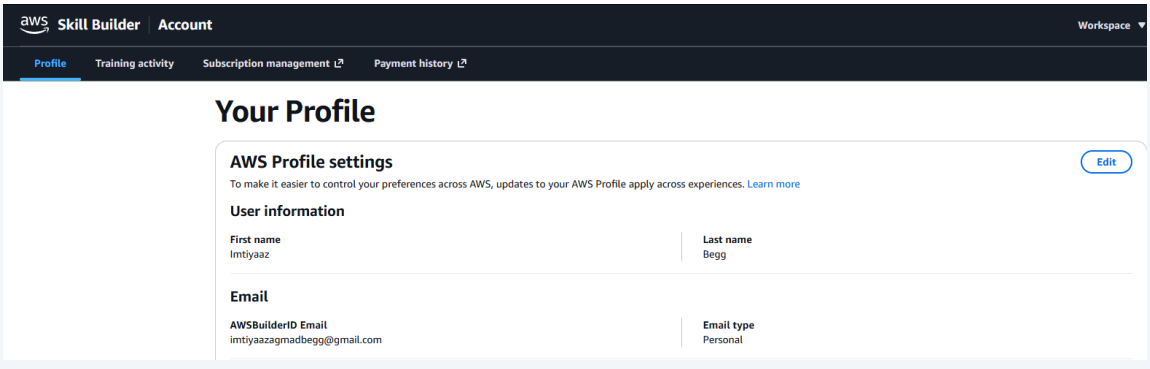
I created a monthly AWS cost budget of USD 5 so that I can keep an eye on spending while completing the practical work. I also set up email alerts at 80% and 100% of the budget. These alerts help me notice when costs are getting close to the limit. They are notifications and do not automatically stop AWS services.





2.3 AWS Skill Builder

I used AWS Skill Builder as my learning platform. It gives me access to AWS learning material that I can use to build my cloud knowledge alongside the practical work.



2.4 Day 1 decision note

My main decisions for the account setup were:

- I kept MFA enabled on the root account because it is the most important account to protect.
- I used a small monthly budget and email alerts so I can monitor costs while learning.

- I chose AWS Skill Builder as my learning platform for AWS training.
- I will avoid leaving resources running if I am not using them.
- I will keep passwords, MFA details and access keys private.

3. Day 2 – Private S3 Data Vault

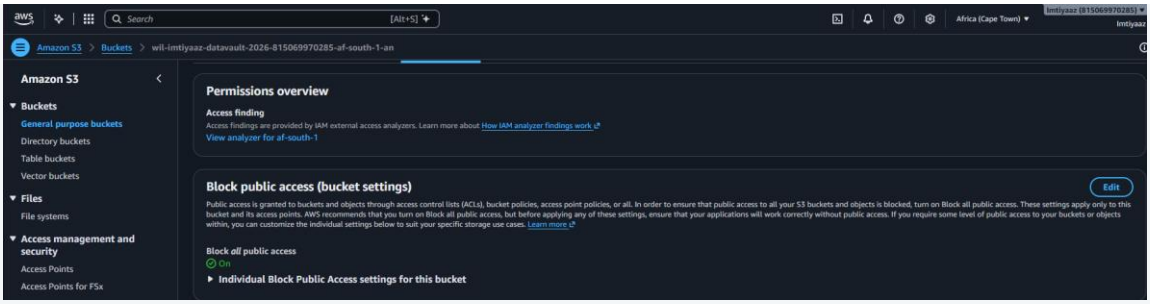
For this task, I created an S3 bucket to use as a small data vault. I kept the bucket private, enabled encryption and versioning, and added the tags required by the sprint. I also uploaded a sample text file so I could test whether versioning would let me recover an earlier copy.

3.1 Bucket details

Bucket name: wil-[yourname]-datavault-2026
Region: Africa (Cape Town) – af-south-1
Bucket type: General purpose

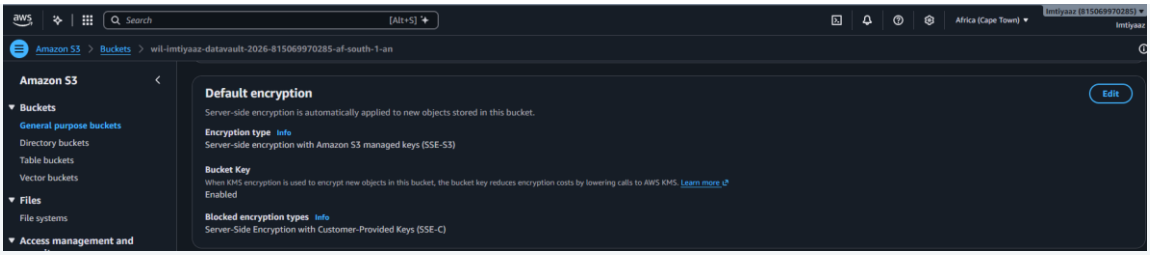
3.2 Public access settings

I kept Block all public access enabled. This means the bucket is not intended to be publicly accessible, which helps prevent files from being exposed by mistake.



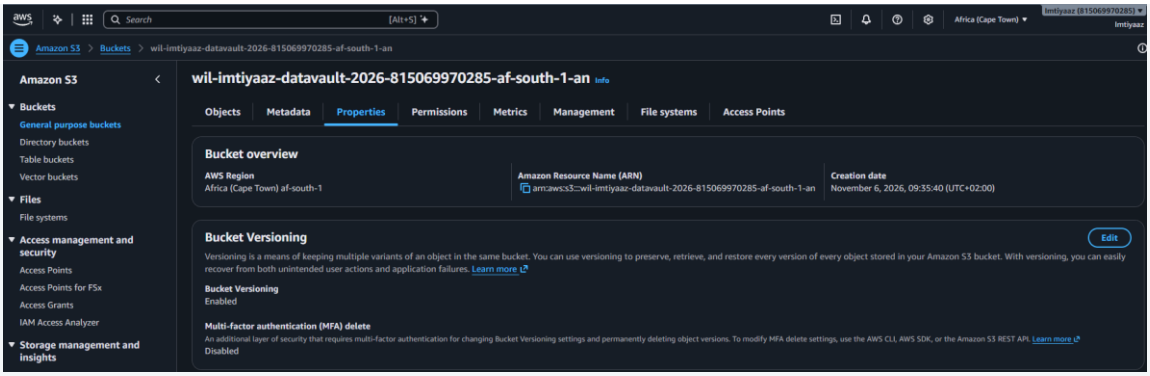
3.3 Default encryption

I enabled default server-side encryption using Amazon S3 managed keys (SSE-S3). This means objects uploaded to the bucket are encrypted by default.



3.4 Bucket versioning

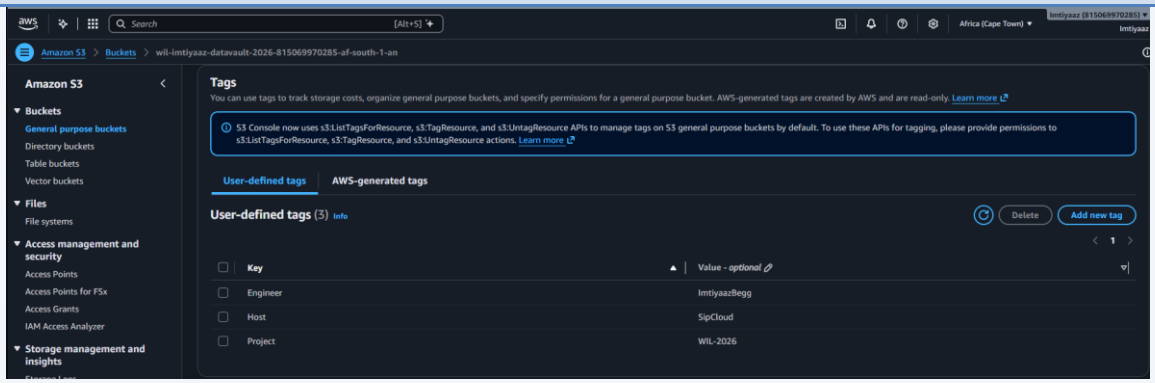
I enabled versioning on the bucket. Versioning keeps older versions of an object when the same file is uploaded again. This is useful if a file is overwritten or changed by mistake.



3.5 Bucket tags

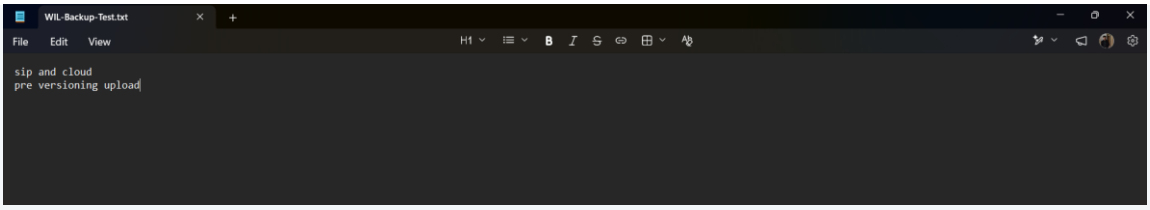
I added the tags below so the bucket can be identified as part of the WIL project.

Key	Value
Project	WIL-2026
Host	SipCloud
Engineer	ImtiyaazBegg



3.6 Sample object upload

I uploaded a sample file called WIL-Backup-Test.txt. I used this file for the restore test so I could check how S3 versioning behaves when an object is overwritten.



4. Day 3 – S3 Restore Test

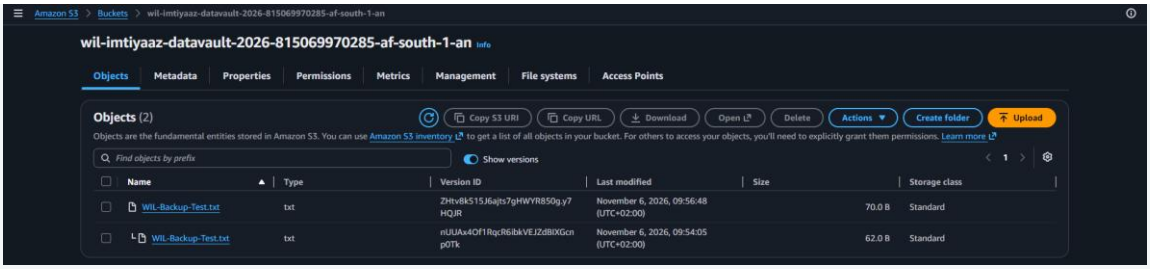
The purpose of the restore test was to check whether I could recover an earlier version of a file after uploading a changed copy with the same name. This is important because versioning can help recover from accidental changes.

4.1 Test procedure

1. I uploaded the original WIL-Backup-Test.txt file to the S3 bucket.
2. I changed the file content and uploaded it again using the same object name.
3. I enabled Show versions in the S3 object list to view the saved versions.
4. I identified the older version containing the original content.
5. I copied or uploaded the older content as the current version and checked the result.

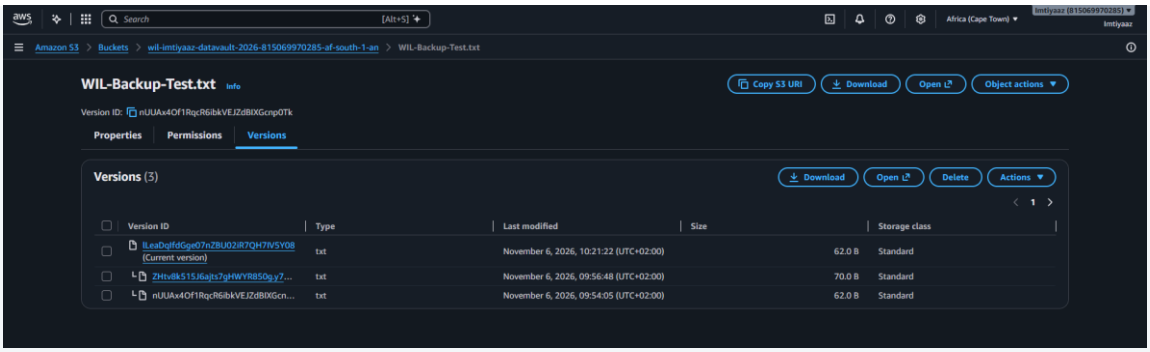
4.2 Version history evidence

The version history should show the different versions of the same object. The screenshot below should be taken after the overwrite and should show the older and newer versions.



4.3 Successful restore evidence

After restoring the earlier content, I checked the latest version of the file. The restore is only complete once the latest version contains the original text again. I will use the screenshot below to prove the result.



4.4 Restore runbook

The steps I followed for recovery are:

6. Open the S3 bucket and locate WIL-Backup-Test.txt.
7. Turn on Show versions.
8. Find the older version that contains the required original content.

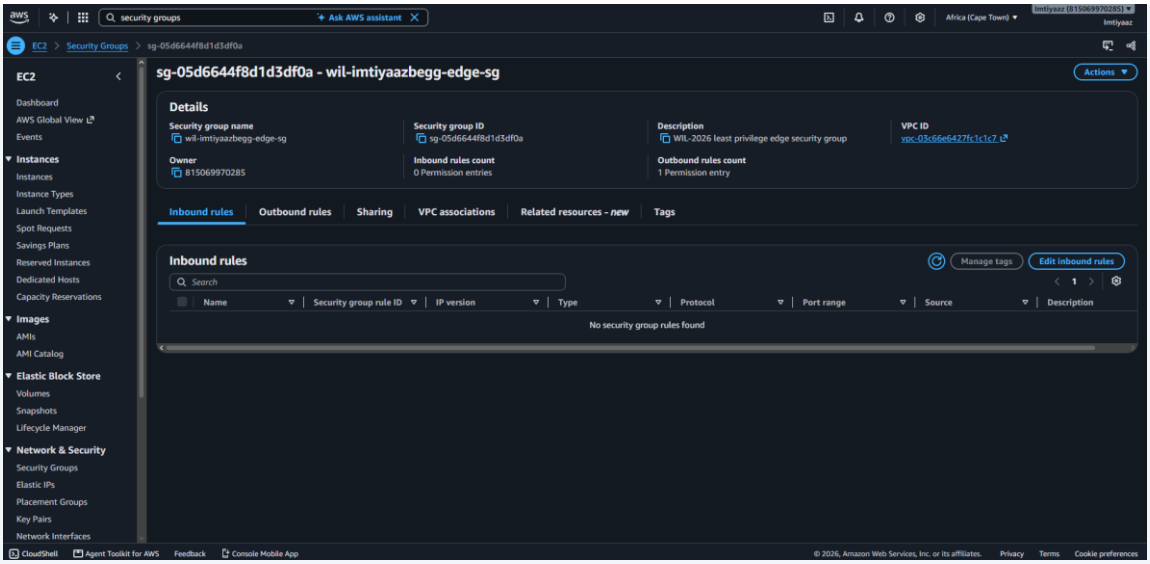
9. Download or copy the older version.
10. Upload or copy it back using the same bucket and object key so it becomes the latest version.
11. Open or download the latest version and check that the original content has been recovered.

Expected result: the original file content is available as the latest version, while the previous versions remain in the bucket history.

5. Day 4 – Security Group and Least Privilege

I created a Security Group called wil-[yourname]-edge-sg. I used the least privilege approach, which means only allowing network access that is actually needed. Since I did not have a workload that required inbound connections for this task, I left the inbound rules empty.

I did not open SSH (port 22) or RDP (port 3389) to 0.0.0.0/0. Leaving these management ports open to the whole internet would create unnecessary risk. The default outbound rule was left unchanged.



5.1 Justification for the rules

- No inbound rules were added because no service in this task needed incoming connections.
- SSH (22) was not opened because remote shell access was not required.
- RDP (3389) was not opened because remote desktop access was not required.
- The default outbound rule was retained for normal outbound communication.

6. Day 5 – NACL Awareness, Decisions and Cost Check

6.1 Security Groups and Network ACLs

A Security Group controls traffic for an AWS resource, while a Network ACL (NACL) controls traffic at the subnet level. They both help manage network access, but they work differently.

Feature	Security Group	Network ACL (NACL)
Level	Resource / network interface	Subnet
State	Stateful	Stateless
Rules	Allow rules	Allow and deny rules
Traffic	Inbound and outbound	Inbound and outbound
Return traffic	Automatically allowed for permitted connections	Must be allowed by separate rules
Rule order	All rules are evaluated together	Rules are checked in number order

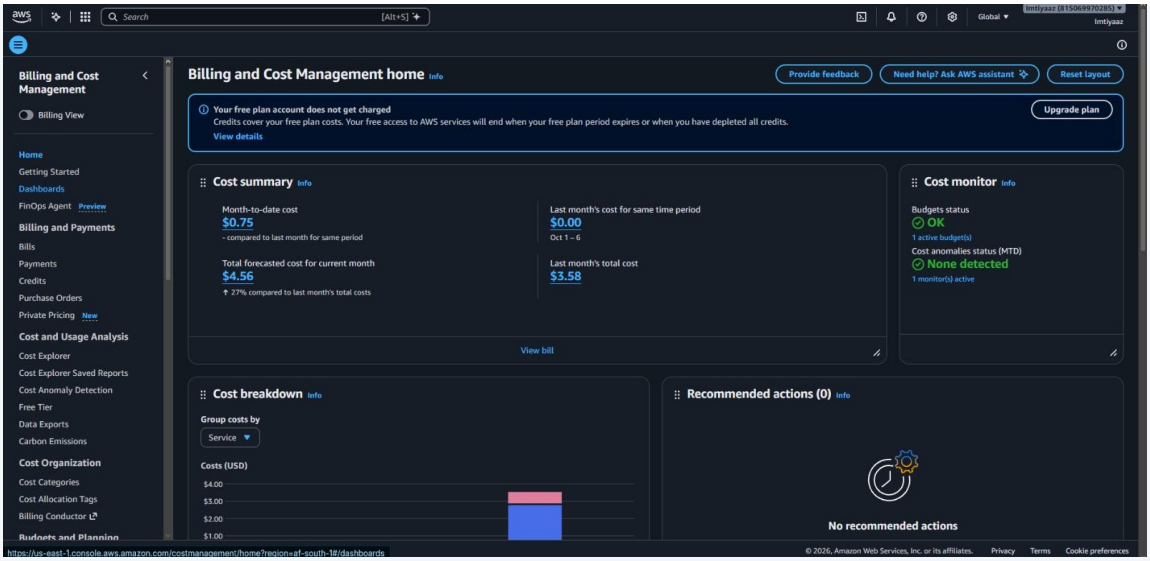
6.2 Day 5 decisions

My main decisions and takeaways from the week were:

- I kept the AWS root account protected with MFA and will use it carefully.
- I created a small monthly budget and alerts to help me keep track of AWS costs.
- I kept the S3 bucket private and enabled encryption and versioning.
- I used versioning as a way to recover an earlier file after an overwrite.
- I chose not to allow unnecessary inbound traffic in the Security Group.
- I learned that Security Groups and NACLs protect different parts of an AWS network, so they can be used as separate layers of security.
- I will check for unused resources and remove them when they are no longer needed to avoid unnecessary costs.

6.3 AWS cost check

I checked AWS Billing and Cost Management to review the costs recorded for the account during the practical work. I will use the screenshot below as evidence of the cost check. The budget alerts are there to notify me if spending approaches the set limit, but I still need to check and manage resources myself.



7. Evidence Index

This index lists the screenshots used to support the work in this document. I will only mark evidence as complete once the screenshot has been captured and checked.

Day	Screenshot file name	What it proves
Day 1	Week1-root-MFA.png	Root MFA enabled
Day 1	Week1-billing-budget.png	Monthly budget
Day 1	Week1-billing-alerts.png	80% and 100% budget alerts
Day 1	Week1-aws-learning-platform.png	Skill Builder dashboard/profile
Day 2	Day2-s3-public-access.png	S3 public access block
Day 2	Day2-s3-encryption.png	S3 default encryption
Day 2	Day2-s3-versioning.png	S3 versioning enabled
Day 2	Day2-s3-tags.png	Required bucket tags
Day 2	Day2-s3-sample-upload.png	Sample object uploaded
Day 3	Day3-s3-version-history.png	Object version history
Day 3	Day3-restore-success.png	Successful restore proof
Day 4	Day4-sg-rules.png	Security Group inbound rules
Day 5	Week1-final-cost-check.png	Final AWS cost check

Note: Screenshots must be inserted in the spaces provided above. Any task that has not been completed or verified should remain clearly marked as outstanding rather than being presented as complete.

8. Week 1 Summary

Overall, this week helped me get more comfortable working in AWS and understanding why security, backups and cost management matter. I learned that creating a resource is only part of the job. It also needs to be configured properly, checked and supported with evidence. I still need to make sure that all screenshots are saved and that the restore test is fully verified before submitting the final version.

Name: Imtiyaaz Agmad Begg

Date: 06/11/2026